

Academic Project Review 1

# Hotel Reservation and Guest Services Management System

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**Name** — Dandigam Ritheek

**Course** — AI-ML

**Institution** — Woxsen University

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Problem Identification

# Why this system is needed

Existing Operational Points

- Difficulty tracking real-time room availability.
- High probability of double-booking during peak periods.
- Inconsistent guest and reservation record keeping.
- Missed or duplicated in-stay service charges.
- Lack of real-time visibility into housekeeping status.
- Manual billing calculations leading to invoice errors.

Problem Statement

"A centralized, automated database system is required to coordinate room reservations, stay management, guest services, housekeeping, billing, and payment processing — while maintaining data consistency and operational speed."

# Who uses the system

| User Role                 | Core Responsibilities & Workflow Scope                                           | Key Permissions                                    |
|---------------------------|----------------------------------------------------------------------------------|----------------------------------------------------|
| Guest                     | Request room reservations, view availability, receive booking & service updates. | Read Availability, Create Reservation              |
| Front Desk / Receptionist | Confirm reservations, check guests in/out, generate invoices.                    | Create/Update Reservation & Stay, Generate Invoice |
| Housekeeping Staff        | View assigned tasks, update room cleaning/maintenance status.                    | Update Housekeeping Task                           |
| Service Staff             | Log guest service requests, record resulting service charges.                    | Create Service Request, Record Charge              |
| Billing / Accounts Staff  | Verify invoices, record and reconcile guest payments.                            | Generate Invoice, Record Payment                   |
| Manager / Admin           | Manage room types & rate plans, view occupancy and revenue reports.              | Full System Access & Analytics                     |

# Main Database Entities

Inventory & pricing

● Room Type

Category defining capacity and nightly rate tier.

● Room

Physical room with number, type, and live status.

● Rate Plan

Pricing plan tied to a room type.

Guest activity

● Guest

Contact details and identity for bookings.

● Reservation

Planned booking — dates, guests, status.

● Stay

Actual check-in / check-out against a reservation.

In-stay services

● Housekeeping Task

Cleaning or maintenance job for a room.

● Service Request

In-stay guest request for a service.

● Service Charge

Billable amount for a fulfilled request.

Billing

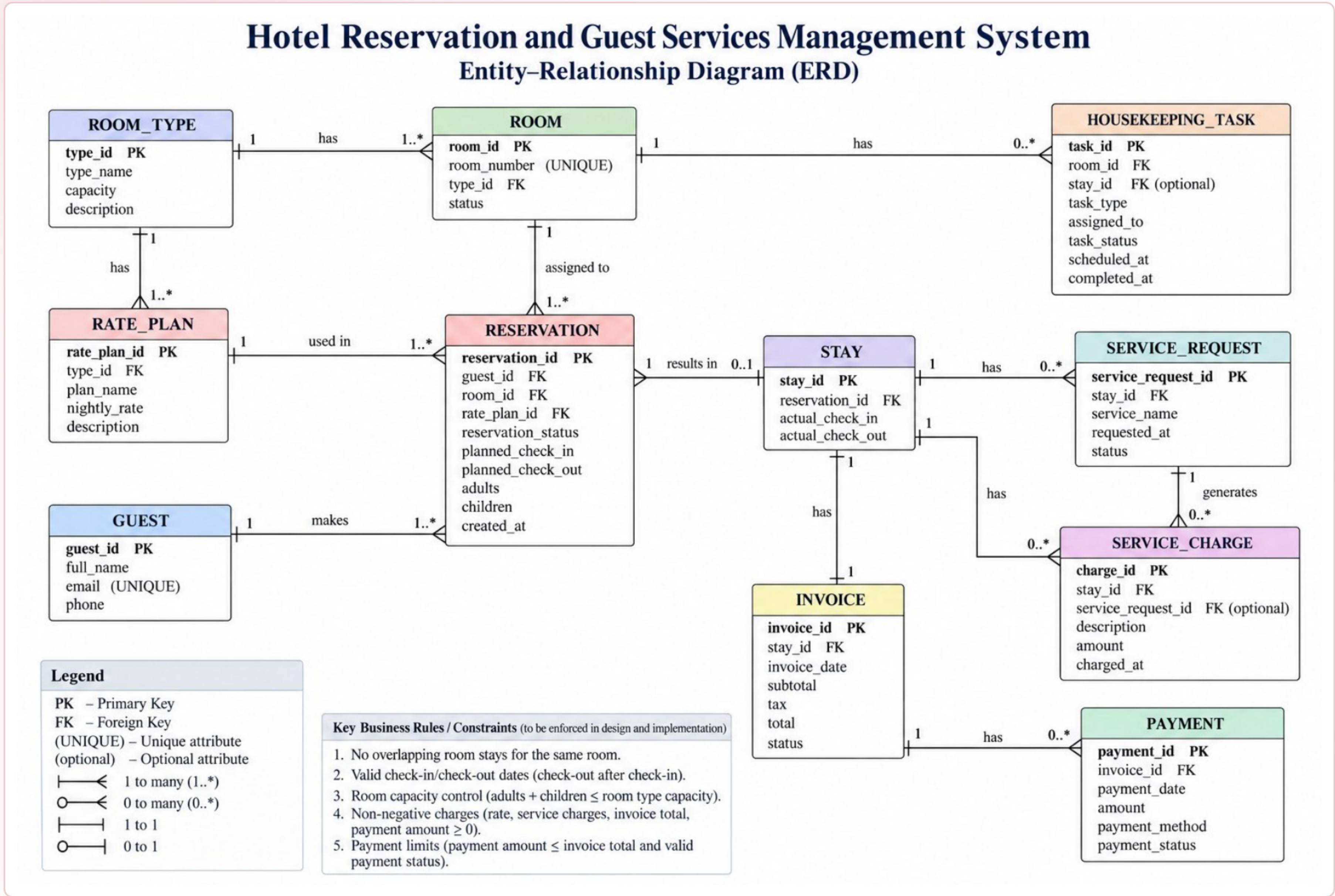
● Invoice

Subtotal, tax, and total for a stay.

● Payment

Amount, method, and status against an invoice.

# Entity–Relationship Model

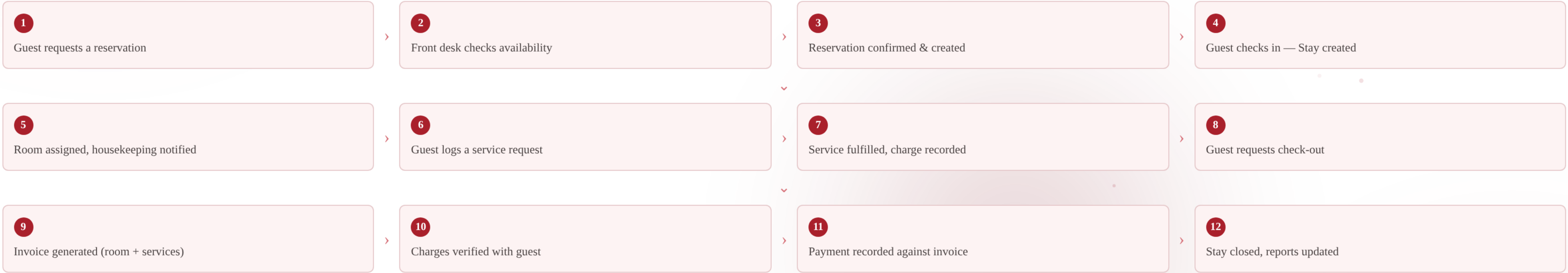


## Core Entity Cardinalities

| Relationship             | Card.  | Meaning                        |
|--------------------------|--------|--------------------------------|
| RoomType → Room          | 1:N    | One type has many rooms        |
| RoomType → RatePlan      | 1:N    | One type has many plans        |
| Guest → Reservation      | 1:N    | One guest, many bookings       |
| Room → Reservation       | 1:N    | Booked at different times      |
| RatePlan → Reservation   | 1:N    | Plan used in many bookings     |
| Reservation → Stay       | 1:0..1 | Realized only on check-in      |
| Room → Housekeeping Task | 1:N    | Many tasks per room            |
| Stay → Service Request   | 1:N    | Many requests per stay         |
| Service Request → Charge | 1:N    | A request can generate charges |
| Stay → Invoice           | 1:1    | One invoice per stay           |
| Invoice → Payment        | 1:N    | Multiple payments allowed      |

Process

# End-to-End Operational Workflow



# Thank You

*Questions & Technical Discussion*